

Time Series Transformer and Scaling Laws

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Review: Scaling Law

Choices of scaling

- Computing Time
- Data Size (number of tokens)
- Model Size (number of parameters)

Constraint: GPUs, training time, budget

Scaling Law (Kaplan et al., 2020)

Describe the law of model performance (or loss) when **other factors are not limited**

Review: The law of power

The power law:

$$L = ax^\alpha + c$$

then

$$\ln L \propto \alpha \ln x$$

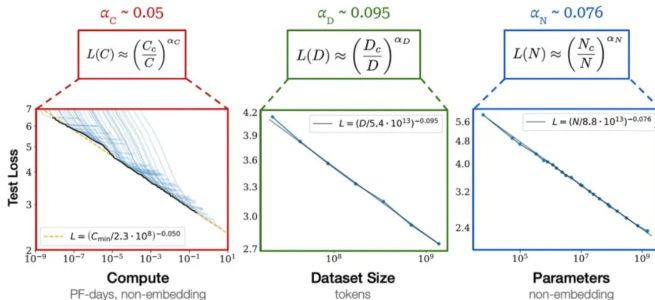


Figure: C,D,N are **potential** limit that can be computed under the scaling law

Review: The scaling low and embedding parameter

Without embedding parameters (token embedding, positional embedding), the scaling law is more clear.

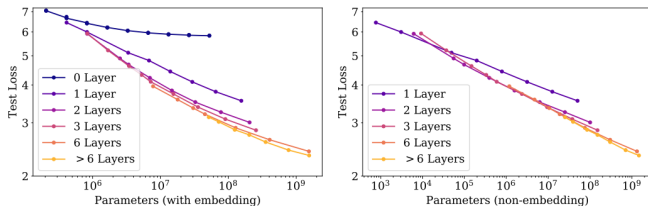
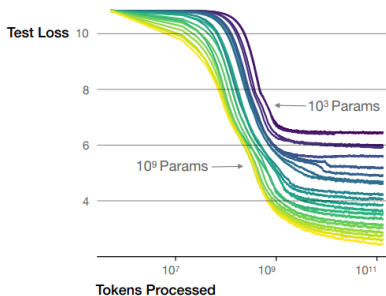


Figure: Loss against parameter (log scale) with embedding (left) and without embedding(right)

Review: When does the model converge

Larger models require **fewer samples** to reach the same performance



The optimal model size grows smoothly with the loss target and compute budget

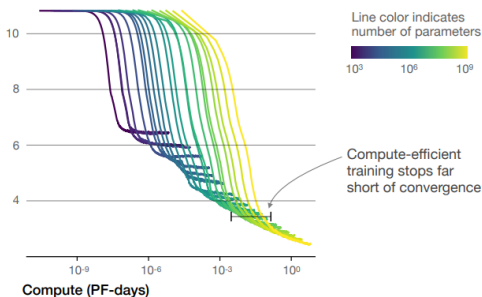


Figure: The token/computing time needed for model to converge

Time Series Transformer

Features of Time Series

- Long Sequence and Model Complexity
- Different "In-Context" Features

Tranformer for Time Series

- LogSparse Transformer (Causal Convolution before Q, K and LogSparse attention $\mathcal{O}(L(\log L)^2)$) (Li, 2019)
- Informer (ProbSparse Self-Attention, $\mathcal{O}(L \log L)$ for memory and computation) (Zhou, 2021)
- Prob Transformer (State Space Models) (Tang, 2021)
- Contformer (Continuous Multi-Head Attention, Neural ODEs) (Chen, 2024)
- MTS Tranformer (Encoder-Only, Unsupervised Learning) (Zerveas, 2021)

Informer and Self-Attention Distillation

- Most queries in time series are "lazy"
- MaxPool after Convolution and Activation

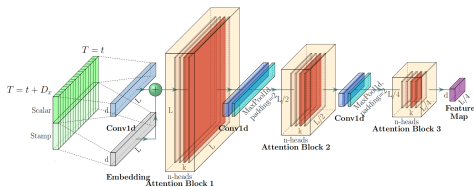
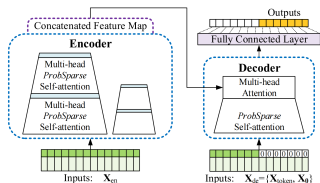


Figure: Query Reduction in Informer

Scaling Law of Time Series Model (Yao et al., 2025)

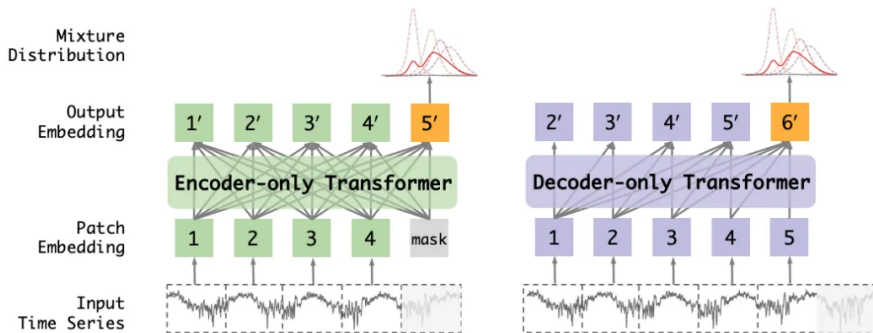


Figure: Two Types of Time Series Transformer from the Author

Scaling Law of Time Series Transformer

Model

- Model Input: Patch Embedding
- Embedding: Rotary Positional Embedding (RoPE) on Query(Q) and Key(K)
- Model Output: Distribution Fitting (Mixture t-distribution assumed)

T-distribution Fitting

$$p(x; \nu, \mu, \tau) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right) \sqrt{\pi\nu\tau}} \left(1 + \frac{1}{\nu} \left(\frac{x - \mu}{\tau}\right)^2\right)^{-\frac{\nu+1}{2}}$$

ν, μ, τ are degree of freedom, location and scale.

Scaling Law: Model Size

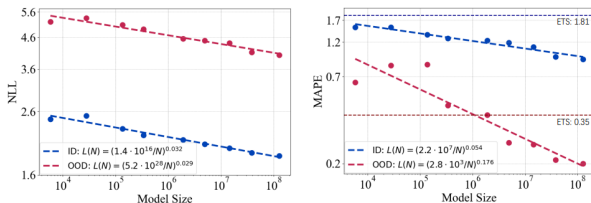


Figure 2: **Parameter Scaling.** The scaling effect of total trainable model parameters on the in-distribution (ID) and out-of-distribution (OOD) forecasting performance, which is evaluated using NLL and MAPE metrics. When evaluated with NLL, both ID and OOD results follow an approximate power law scaling with parameter count, exhibiting consistent trends across different data distributions. The blue and red horizontal dashed lines represent the baselines of the exponential smoothing (ETS) forecasting method.

Scaling Laws of Dataset Size and Computing

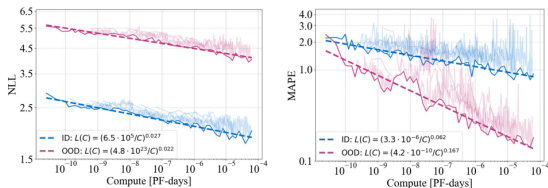


Figure 3: **Compute Scaling.** The computation scaling results indicate that model performance scales approximately according to a power law with increasing compute, consistent across both ID and OOD scenarios. The ID and OOD results illustrate that there is an lower bound for loss and MAPE on both test data under a given computational budget.

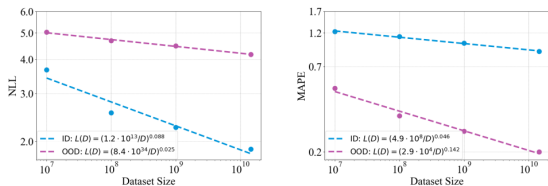


Figure 4: **Data Scaling.** The blue and red plots illustrate how data volume affects the ID and OOD forecasting performance of encoder-only Transformers, evaluated using NLL and MAPE metrics. The results indicate that in both scenarios, model performance scales approximately as a power law with data volume.

In Domain vs Out of Domain, Encoder vs Decoder

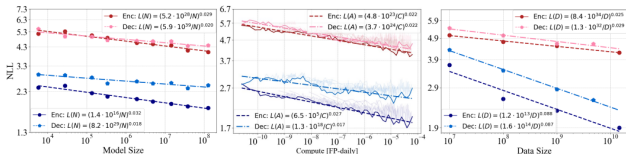


Figure 5: Scaling Laws of Encoder-only vs. Decoder-only Transformer. This figure presents a comparison of scaling behaviors on NLL between encoder-only and decoder-only Transformer across three different axes: number of parameters, compute, and dataset size. Overall, both models exhibit similar scalability patterns with respect to model parameters, computation and dataset sizes across **ID** and **OOD** data, but differ in ID performance.

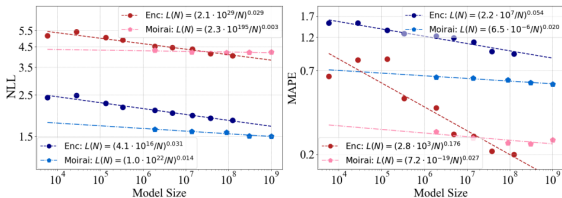


Figure 6: Scaling Laws of Encoder-only Transformer vs. Moirai. This figure compares parameter scaling between the encoder-only Transformer and Moirai. While Moirai demonstrates notable improvements on **ID** test data compared to the baseline, certain design choices may hinder its scalability on **OOD** data.

Application of Scaling Law

Main Conclusion of Scaling Law

- Increase computing time, dataset size, and model parameters **at the same time**

Interesting Findings about Scaling Law

- Train large model on small dataset, and stop training before converge (without the considering inference cost)
- Different Scaling Speed: 8 times Model Size vs 5 times Data Size

Applications

- **Predict the model performance before training (consider the budget)**
- Infer the performance of larger models with scaling law